

AI Software Project Management Tools

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Chapter 7

AI Software Project Management Tools



Abstract This chapter explores the most popular AI-powered tools used in software project management, focusing on their functionalities and applications in real-world scenarios. It provides an overview of how these tools enhance project planning, monitoring, and execution. Additionally, the chapter analyzes the benefits, such as increased efficiency and automation, as well as the challenges, including implementation costs and data quality, that come with integrating AI into software project management tools.

Keywords AI tools in project management · Software project planning tools · Benefits of AI in project management · Challenges of AI in project management

Software project management plays a crucial role in software companies, enabling them to achieve their goals related to the execution of software projects. There is always a certain number of projects that can become unsuccessful, either due to the way they are managed or due to their end result and achievements. Since the success of any initiated project has a direct financial impact on a company's operations, they are compelled to undertake various activities to ensure the success of their projects [1]. One such activity is the implementation of tools that enable the prediction of project flow and completion based on input parameters, as well as the identification of critical elements that have a direct impact on project outcomes. Lately, these tools increasingly rely on the advancements of artificial intelligence. Artificial intelligence has been mentioned in research papers analyzing its impact on project management for almost three decades. However, only in recent years has the influence of artificial intelligence advancements on project management become clearly visible. Currently, organizations are actively incorporating different process automations and data integrations into their operations. However, it is important to note that artificial intelligence holds tremendous potential for enhancing project management [2]. The implementation of specific AI tools greatly facilitates and empowers a range of crucial capabilities in project management, such as intelligent decision-making, predictive analytics, automated task management, resource optimization, and more. Different artificial intelligence tools can analyze historical project data, identify patterns, and

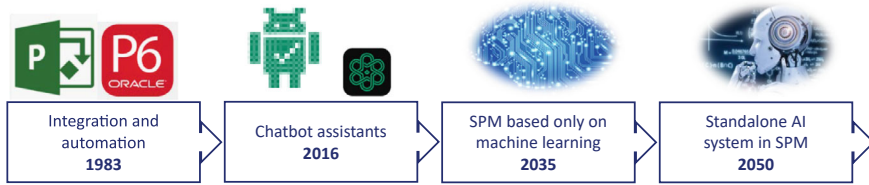


Fig. 7.1 An overview of the development of artificial intelligence in project management thus far and assumptions for future development [5]

predict potential risks and their impact on the project. Moreover, these tools can provide recommendations for risk mitigation and project outcome improvement. Additionally, AI software tools can identify key factors that directly influence project outcomes, such as the identification of critical tasks, resources, or risks that require special attention and management [3]. The application of artificial intelligence tools in project management depends on the specific needs and objectives of the project and can involve various functionalities and applications. In addition to prediction and identification, artificial intelligence tools can also be used for data analysis, resource optimization, improving user experience, and similar purposes [4]. The application of artificial intelligence in software project management is evolving in several stages: starting with the use of chatbot assistants for project management, then utilizing machine learning algorithms in software project management tools, and finally, the replacement of the entire project management system with a fully autonomous artificial intelligence system is expected [5]. An illustration of this artificial intelligence evolution is shown in Fig. 7.1.

The first phase of tool development and application involves software for integrating and automating task activities in software project management. The most widespread and commonly used tools in this initial phase include Microsoft Project and Primavera developed by Oracle. According to the authors of this classification [6], the second phase of AI tool application involves the use of chatbot assistants. In recent years, they have been applied and used to record meetings and provide reminders to team members, tracking their performance on one hand and enabling project managers to monitor all activities. Examples of chatbot usage and application are integrated into Slack-Fireflies.ai [7]. The third phase of AI development in project management involves the application of machine learning. The authors state that the ML algorithms, which form the core of AI, provide the possibility of predictive and corrective analysis. In practice, this would mean that project managers have access to data that could assist them in decision-making during the planning and resource management phases, within certain parameters and constraints. In addition to addressing fundamental ML problems, it would enable risk minimization to successfully complete the project. Predictions suggest that within less than 10 years, AI will be capable of proposing project schedules based on the analysis of a large amount of historical data, depending on all available project resources. Moreover,

new tools will be able to alert project managers to any risks or unforeseen circumstances that may arise during project implementation, which the manager may not be able to detect in a timely manner due to the vast amount of data and information [8]. The final phase in the application of artificial intelligence in software project management in the further future, around 20 years from now, would involve autonomous systems with minimal human intervention. It is believed that autonomous systems will be able to independently analyze, predict, and draw conclusions about the necessary activities for controlled and successful project completion by inputting a very small number of parameters [6, 7]. Although the application of artificial intelligence will enable the automation of many tasks performed by project managers, there are certain human qualities that artificial intelligence currently does not possess and will not possess in the near future. These are the human leadership factors that involve effective team leadership and mentoring of project team members, as well as the way of communicating with stakeholders. These are areas where human interaction, common sense reasoning, empathy, and a certain way of connecting with people are necessary, which machines are unable to achieve [2]. This implies that the role of the project manager will not disappear with the emergence of artificial intelligence. Artificial intelligence will become a reliable partner to the project manager, performing repetitive tasks that often take up more than half of the project managers' working time, allowing them to focus on other areas of project management. The project manager will need to expand their knowledge to include digital areas, but they will have more time that they can utilize efficiently. Artificial intelligence does not threaten the existence of the project manager position; instead, it adds value to it because now the person managing the project can focus on more important tasks such as planning, strategic objectives, and other underlying processes [3]. They have more time to dedicate to their team, work on improving the work environment, and address everything the team needs, which will make the team feel more valued, empowered, confident, and therefore more efficient. The project managers can then focus on tasks they consider essential, as they have not had time to address them due to handling a large amount of data. The position of the project manager is changing and gaining breadth, as they now become a leader, not just a manager [9].

The most commonly used tools in Software Project Management will be presented below. In Sect. 7.1, while Sect. 7.2 will provide the Benefits and challenges in using artificial intelligence in project management tools.

7.1 Most Popular Tools Used in Software Project Management

1. **TensorFlow** (www.tensorflow.org) is a widely used framework for machine learning and deep learning with applications in various areas of artificial intelligence, including software project management. It provides developers with the ability to develop, train, and deploy various artificial intelligence models.

In the context of software project management, TensorFlow has several useful applications [10, 11].

The first application is data analysis, where TensorFlow offers a rich set of libraries and tools for analyzing data. This can be crucial in software management projects, enabling exploratory data analysis, pattern recognition, segmentation, and other analytical data processing tasks.

The second application is predictive analytics, where TensorFlow is used to develop and train machine learning models that can predict specific aspects of a software project. For example, it can be applied to predict project deadlines, assess risks, identify errors, or optimize performance.

The third application is natural language processing. TensorFlow has a module for natural language processing that allows analysis and processing of textual data. This is particularly useful in projects that require the analysis and understanding of user requirements, communication, or processing of large amounts of textual data.

The fourth application is resource optimization. TensorFlow can be used for resource optimization in software projects, such as optimizing resource scheduling, finding the most efficient algorithms, or optimizing the performance of software systems.

Finally, TensorFlow can also be applied to enhance the user experience of software products. For example, it can be used to develop models that improve personalized recommendations or for image and audio recognition and processing in user interactions [11] (Fig. 7.2).

TensorFlow offers a comprehensive suite of tools and functionalities that empower the development and execution of various machine learning models, including neural networks. This versatile framework supports multiple platforms, such as GPUs, distributed computing, and mobile devices, allowing for efficient deployment and scalability. In addition to the core TensorFlow framework, the vibrant TensorFlow community has developed additional packages and extensions that expand its application across diverse domains. For instance, TensorFlow.js facilitates machine learning execution within web browsers, TensorFlow Lite caters to resource-constrained devices, and TensorFlow Extended (TFX) focuses on the development, implementation, and maintenance of production-ready machine learning models [12]. TensorFlow, being an open-source software framework, enables users to customize, utilize, and distribute its source code under the Apache 2.0 license. This flexibility, combined with its support for various platforms and an engaged community, has contributed to TensorFlow's popularity in numerous artificial intelligence projects, including software project management. Among the TensorFlow extensions, TensorFlow Lite

Fig. 7.2 TensorFlow logo



Fig. 7.3 MS project logo

Micro (TFLM) stands out as an open framework specifically designed for machine learning inference on embedded systems. TFLM addresses the efficiency requirements that stem from resource constraints in embedded systems and tackles the challenges of fragmentation, hindering cross-platform interoperability. Through its unique interpreter-based approach, TFLM overcomes these challenges, providing flexibility and enabling the application of deep learning on a wide range of embedded systems. By extending the reach of machine learning into embedded systems, TFLM makes a significant contribution, especially in the context of software project management. It is tailor-made to efficiently execute machine learning models on devices with limited memory, such as embedded systems with only a few kilobytes of memory. The core objective of TFLM is to address the challenges posed by hardware heterogeneity, fragmented ecosystems, limited software functionalities, and severe resource constraints commonly encountered in embedded systems [13] (Fig. 7.3).

2. **Microsoft Project** (www.microsoftproject.com) is a project management software developed by Microsoft. This tool provides comprehensive support to users in planning, tracking, and managing projects of various sizes and complexities [14]. Its advanced features enable users to meticulously plan resources, schedule tasks, track progress, analyze risks, and generate reports.

One of the key features of Microsoft Project is the ability to plan projects. Users can create detailed plans that encompass all tasks, timelines, dependencies, resources, and schedules. This includes creating Gantt charts and defining timelines for each task, enabling a clear understanding of the project schedule. Resource scheduling is another important feature of Microsoft Project. Users can manage resources such as people, materials, and equipment, and assign them to tasks within the project. The tool provides the ability to track and manage resource capacity and schedules to ensure optimal resource utilization and avoid overloads or shortages. Project progress tracking is also facilitated with Microsoft Project [15]. Users can enter actual task completion data, modify schedules, or add new tasks to reflect the real state of the project. Based on the entered data, the tool automatically updates charts, displays deviations from the plan, and generates progress reports. This allows users to stay up to date with the project's progress and take necessary corrective actions if needed.

Risk analysis is another important capability provided by Microsoft Project. Users can identify, assess, and manage project risks. They can add risks to the project, assign them probabilities and impacts, and conduct analysis to assess overall risk and take appropriate measures to mitigate risks [16]. This feature allows users to be proactive

in risk management and minimize potential issues that can impact project success. Moreover, Microsoft Project provides robust support to users in project risk analysis. It allows them to identify, assess, and manage risks by adding risks to the project, assigning probabilities and impacts, and conducting analysis to assess overall risk. Based on this information, users can take appropriate measures to mitigate risks, increasing the chances of successful project execution.

Report generation is another valuable feature of Microsoft Project. The tool enables users to generate various reports such as Gantt charts, resource tables, cost reports, and project progress reports [17]. These reports provide users with detailed insights into key aspects of the project, enable progress tracking, and facilitate communication with relevant project stakeholders. This helps users stay well-informed and make informed decisions for the project's further advancement.

Microsoft Project is commercial software, and users need to purchase a license or subscription according to Microsoft's terms to use it. This license grants users access to all features and functionalities of the tool and ensures lawful and supported usage [18].

3. **Wrike** (www.wrike.com) is a project management and team collaboration tool that enables organization, tracking, and management of tasks, projects, and collaboration within teams and organizations. It is designed to improve productivity, transparency, and efficiency in software project management [19] (Fig. 7.4).

Wrike is a comprehensive project management tool that provides various capabilities for organizing, tracking, and collaborating on projects. Its functionalities allow for the creation of project plans, the definition of deadlines, and the tracking of project progress. Additionally, it enables team collaboration through a shared workspace where teams can share files, communicate through comments, send notifications, and assign tasks. This integration facilitates team communication and collaboration on projects. Wrike also provides resource management capabilities, task scheduling, and productivity tracking. Teams can monitor work capacities, assign tasks based on resource availability, and track their productivity. This functionality assists in optimal resource allocation and prevents overload [20].

The tool also allows for tracking time spent on tasks and monitoring project costs. Users can track time spent on each task, generate time tracking reports, and monitor costs in real-time. This enables better project budget management and cost tracking. One of the key advantages of Wrike is the ability to generate project and team productivity reports. Users can track key indicators of project efficiency, analyze progress and team productivity data, and generate charts and visual representations. This provides better insights into project performance and enables informed decision-making. Wrike is not open-source software but a commercial product used through a subscription based on the terms and plans offered by Wrike Inc [21].

Fig. 7.4 Wrike logo



4. **Monday** (www.monday.com) is a project management platform that provides a range of functionalities and features to help address project management challenges. The Monday tool allows teams to centralize all project-related information in one place. This includes tasks, deadlines, documents, notes, comments, and any other relevant details [22]. This centralization helps reduce data fragmentation and facilitates access to all necessary information. The platform provides a visual representation of project tasks, their status, and priority points. Teams can clearly see who is responsible for which task, when the deadline is, what the priorities are, and how much progress has been made. This helps in efficient resource allocation and project tracking. Monday supports team collaboration in multiple ways. Teams can assign tasks to team members, communicate through comments, share files, and provide feedback. This functionality improves communication and collaboration within the team, helping address issues related to lack of transparency and coordination. This tool enables the automation of routine and repetitive processes [23]. Teams can set rules that automatically trigger certain actions based on specific conditions or events. For example, it can automatically assign tasks, send reminders, or update task status. Process automation helps reduce human error, improves efficiency, and saves time. Monday allows for generating various types of reports and project progress views. Teams can track statistics, metrics, deadlines, costs, and other relevant data. These reports provide insights into project performance and help identify issues and make informed decisions. Monday provides flexibility and adaptability to cater to the specific needs of projects and teams [24]. Monday is commercial software, and a license or subscription is required to use it according to company's terms (Fig. 7.5).
5. **Jira** (www.jira.com) is a popular project management tool developed by Atlassian. It is designed for tracking, managing, and collaborating on tasks, projects, and teams within organizations. Jira is commonly used in agile software development environments but can be adapted to other types of projects as well [25]. Jira is a versatile project management tool used in various industry sectors. This tool allows for task definition, assignment to team members, progress tracking, and deadline management. Project managers can easily monitor the project workflow and identify tasks that require attention. Jira provides support for agile methodologies such as Scrum and Kanban [26]. Teams can utilize sprints, backlogs, task boards, and burndown charts to organize work and achieve project goals. The tool enables team collaboration through a shared workspace. Team members can communicate, share files, comment on tasks, and provide feedback in one place. This facilitates communication and increases transparency in work. Jira allows for resource tracking and task assignment based on availability and team members' skills. Project managers can optimize resource allocation and avoid overloading specific team members. Jira is often used for tracking and managing

Fig. 7.5 Monday logo



Fig. 7.6 Jira logo

bugs, user issues, and other problems that arise during the project [27]. Teams can assign priorities, allocate tasks for issue resolution, and track their resolution (Fig. 7.6).

Jira enables storing and managing project documentation in one place. Teams can share, version, access, and update documents through Jira. This facilitates organization and access to relevant project-related documents. This tool allows project managers to plan capacities and resources based on project requirements. Teams can determine resource availability, assess their overall capacity, and plan tasks and schedules accordingly. Jira also supports change management processes in projects [28]. Teams can track and manage change requests by monitoring their status, priorities, and implementation to ensure controlled changes in the project. It is often used in software development as a central tool that integrates different phases of the development process. It can be integrated with version control tools, issue tracking, test automation, and similar tools, enabling tracking and management of all aspects of the development cycle. Additionally, Jira provides project portfolio management capabilities, allowing for an overview of all projects at the organizational level. This includes tracking progress, resources, priorities, and financial aspects of each project to make informed decisions about investments and resources. Jira allows teams to track time spent on tasks and projects and record costs. Working hours can be created and tracked, reports on time spent can be generated, and cost analysis can be performed to better manage the project budget. The tool is specifically tailored for project management using agile methodologies such as Scrum and Kanban [29]. It provides features for tracking agile backlogs, sprints, burn-down charts, task management, and team collaboration. Jira enables tracking project requirements and managing their lifecycle. Teams can create, prioritize, assign, and track requirements to ensure that customer and stakeholder needs are met. This tool is often used for incident tracking and user support. Teams can collect, prioritize, and resolve issues and user inquiries through Jira. This allows for better time management of response, monitoring support quality, and ensuring customer satisfaction. Jira can be integrated with other tools and services such as code versioning, test automation, task management tools, and more. This integration enables better data and process synchronization between different tools, resulting in more efficient project management [30]. Lastly, Jira provides the ability to generate various reports and metrics to track project progress, team performance, work quality, and other relevant parameters. Reports can be customized to specific project and organizational needs and used for making informed decisions.

6. **Trello** (www.trello.com) is a project management and team collaboration tool that utilizes the concept of boards and cards to organize tasks and facilitate collaboration within teams. It is designed to simplify task tracking and management,

Fig. 7.7 Trello logo [31]

project organization, and team activities. The concept of “boards” and “cards” allows users to organize tasks, information, and project resources in an intuitive manner [31]. Users can create lists and cards and move them between different lists to represent progress and task status. Each card in Trello can represent an individual task or a task within a project. Users can add task details, deadlines, assignees, and comments to the cards [32]. Trello enables easy tracking of task progress, moving cards between lists to indicate task status (e.g., “In Progress,” “Completed”), and assigning priorities. Users can add team members to cards and lists, allowing them to view, comment on, and contribute to tasks. Trello supports assigning responsibilities for tasks and designating team members accountable for specific activities. This platform enables users to attach files and add them to cards, facilitating easy sharing with team members. Trello has the ability to streamline resource distribution, including important documents, images, videos, and other project materials. Moreover, this tool allows for the integration of add-ons and connects with other tools, expanding its resource management capabilities. It offers a comprehensive project overview, ensuring that all team members have visibility into the current status, progress, and assigned tasks. This enhances project transparency and fosters seamless communication within the team [33] (Fig. 7.7).

Trello enables integration with various tools and platforms such as Google Drive, Dropbox, Slack, Jira [34] and many others. This facilitates the exchange of data and information between different tools that could be used for software project management. It offers the ability to add deadlines to tasks and track time spent on tasks. Users can set reminders and alerts to ensure tasks are completed within the set deadlines [35]. Furthermore, they have the capability to generate reports on projects and team productivity. Additionally, users can track statistics such as the number of completed tasks, time spent on projects, and other key performance indicators. These reports assist in assessing project effectiveness and identifying areas for improvement. The tool is available as a mobile application for iOS and Android devices [36]. This allows users to access and collaborate on projects even while on the go. Trello is not open-source software. The source code of Trello is not publicly available and cannot be freely modified or customized. It is a commercial product that users can access through a subscription based on the terms and plans offered by Atlassian, the company that owns Trello.

7. **Asana** (www.asana.com) is a project management and team collaboration tool. It is designed to help organize, track, and manage tasks, projects, and collaboration within teams and organizations. Asana assists teams in staying organized, aligned, and productive [37] (Fig. 7.8).

Fig. 7.8 Asana logo

Asana enables users to create project boards that represent projects and contain various tasks and activities. It is possible to organize tasks into sections, add descriptions, assign tasks to team members, and set deadlines. This facilitates tracking project progress and ensures tasks are completed on time. This tool allows teams to collaborate on projects through shared boards. Team members can comment on tasks, ask questions, share files, and keep each other informed of progress. This functionality facilitates communication and collaboration within the team, which is crucial for successful project execution [19]. Asana allows tracking and management of resources in a project. Additionally, assigning tasks to specific team members, tracking workloads, and allocating tasks based on resource availability is also possible in this tool. This helps optimize resource allocation and avoid overloading. It enables setting deadlines for tasks and project goals [38]. Alerts and reminders can be set to stay informed about upcoming deadlines and ensure tasks are completed on time. This is particularly useful for tracking projects with time-sensitive tasks. Asana provides a clear overview of project progress [39]. It is possible to track completed tasks, tasks in progress, and tasks that need to be done. Additionally, Asana offers project reporting capabilities, allowing performance analysis, tracking key performance indicators, and making informed decisions [40]. This tool can be integrated with other tools and platforms such as Gmail, Slack, Google Drive, and many others. These integrations facilitate data synchronization and collaboration. Asana is not open-source software. The source code of Asana is not publicly available and cannot be freely modified or customized. Asana is a commercial product that users can access through a subscription based on specific terms [41].

8. **Testlio** (www.testlio.com) is a software testing management platform that provides users with external Quality Assurance (QA) support. It is designed for companies and teams looking to improve the quality of their software and ensure that their products or services meet high standards [42] (Fig. 7.9).

Testlio allows teams to create test plans, execute tests, and track software bugs. It is possible to define test scenarios, run tests on various platforms and devices, and

Fig. 7.9 Testlio logo

track and document the bugs found. It facilitates collaboration among teams for software testing. Testers can be assigned to tasks, exchange information and comments, and track testing progress [43]. Additionally, teams can communicate through an internal chat system and share documentation and resources. Testlio supports test automation using different tools and techniques. Automated test scripts and tools can be utilized to expedite the testing process and guarantee consistent results. This tool can be integrated with popular bug tracking tools such as Jira, Asana, Bugzilla [44] and others. This enables smooth synchronization between testing and bug tracking, making project task management and status tracking easier. Testlio provides advanced reporting and analytics capabilities. It is possible to generate reports on testing progress, software quality, bugs found, and other relevant metrics. These data help to understand the project's status, identify potential issues, and make informed decisions [45]. This tool is flexible and can be customized to fit the project or team's specific needs. Custom workflows can be defined, priorities can be set, user access can be managed, and the tool can be configured according to individual preferences. Testlio allows creating testing questionnaires, enabling the asking of specific questions to testers about software functionality or user experience. This helps gather valuable feedback and understand the user's perspective. It enables the organization of test frameworks and libraries for faster and more efficient test execution. Tests can be grouped by categories, reusable testing components can be created, and teams' access to relevant tests can be facilitated [46]. Testlio offers version control capabilities for software. Tracking and managing different versions of the software being tested is possible, enabling close monitoring of changes and identification of version-specific issues. The tool provides support for software localization testing, allowing testing of the software in different languages and verification of correct implementation of localization aspects. This is particularly useful for teams developing global products with multilingual support. Additionally, this tool supports security testing and penetration testing of software. Testlio is not open-source software. It is a commercial platform provided as a service by the company. The Testlio source code is not publicly available and cannot be freely modified or customized [47].

9. **Click Up** (www.clickup.com) is a comprehensive project management platform that provides various tools and functionalities for managing and tracking projects. Project management tools like ClickUp enable teams to organize, collaborate, and coordinate work tasks and projects in an efficient manner [48] (Fig. 7.10).

ClickUp is a comprehensive project management platform that provides various tools and functionalities for managing and tracking projects. Software Project Management tools like ClickUp enable teams to organize, collaborate, and coordinate work tasks and projects efficiently. This tool allows the creation of a hierarchical

Fig. 7.10 Click up logo



structure of projects and subprojects. Users can add tasks, assign them to teams and individuals, define deadlines, set priorities, and track progress [49]. Different labels and filters can be used to organize tasks based on various criteria. This tool facilitates communication, document sharing, and task commenting within the platform. It can be used for idea sharing, information exchange, discussion tracking, and decision making. Integration options with communication tools like Slack further enhance collaboration. It provides a clear interface for tracking project progress. Completed tasks, remaining tasks, assigned resources, deadlines, and more can be monitored. Additionally, project progress reports can be generated to gain a clear understanding of the project status. The platform enables file storage and sharing within projects, allowing teams to download, organize, and share relevant documents, images, videos, and other materials. Integration with document collaboration tools like Google Docs or Microsoft Office Online is possible [50]. It provides tools for project resource planning and tracking, including the assignment of resources to tasks, tracking their availability and workload, and optimizing resource allocation for improved efficiency and productivity. Moreover, it supports Scrum and Kanban methods for agile project management, enabling the creation of sprints, defining backlogs, tracking task progress, and using Kanban boards for visual workflow tracking and management. Time spent on tasks can be tracked, priorities can be set, and obstacles or blockages can be identified. Priorities and deadlines can be added to tasks, clearly defining urgency and execution time, thereby facilitating focus on key tasks and maintaining project goal alignment [51]. The tool enables the creation of Gantt charts, showcasing the project schedule, task deadlines, and their interdependencies. This visual representation aids in understanding the project timeline and identifying crucial paths or delays. Furthermore, it offers a range of features designed to support agile software development methodologies. Agile boards (such as Kanban or Scrum boards) can be used to track task status, manage sprints, monitor pull requests, and other agile processes. ClickUp integrates with popular version control and software development tools like GitHub, Bitbucket, and GitLab. This enables automatic task status updates based on code changes or pull requests, fostering better synergy between development teams and project management. This tool offers task automation capabilities through user-defined rules [52]. Automation can be set up to generate notifications, change status, assign tasks, and perform other actions based on specific events or conditions. ClickUp enables tracking of time spent on tasks through integrations with time tracking tools. This feature helps monitor team productivity, estimate task completion time, and generate time tracking reports. Additionally, the tool provides visual displays of resource availability, including calendars and schedule overviews, facilitating resource planning, coordination, and workload tracking. It also offers bug tracking capabilities within the development process, allowing users to create and track bug reports, assign them to responsible individuals, monitor their status and progress, and generate reports on resolved issues.

The platform offers also a wide range of reporting options to monitor project progress and analyze performance. Reports can be generated on tasks' time allocation, adherence to deadlines, resource distribution, and more. Advanced analytics

tools provided by ClickUp enable in-depth examination of project data. Team statistics can be tracked, trends analyzed, areas for improvement identified, and informed decisions made based on the collected data [52]. For tracking task time and team productivity, features to monitor the time spent on individual tasks, generate time tracking reports, and identify opportunities for enhancing efficiency are available. The tool facilitates portfolio management by allowing the tracking of multiple projects in a single location. This aids in obtaining an overview of all projects, including their status, priority, and allocated resources [53]. ClickUp's mobile application enables access to and management of projects using mobile devices, ensuring connectivity and updates even while away from a computer. The platform offers diverse functionalities and capabilities, continually evolving and updating to provide an optimal software project management experience [54].

7.2 Benefits and Challenges in Using AI in SPM Tools

The selection of a tool that will be suitable for a particular organization or project depends primarily on the type and duration of the project, risk prediction capabilities, team size, characteristics, and location, communication methods, integration options, project overview features offered, user progress notifications, task reminders, as well as the cost of the tool. It is also important whether the chosen tool provides training to each user before use, offers additional options for budgeting, invoicing, and sending client proposals. Security features are also essential, such as working with encrypted data, network security on which the tool operates, and a plan for recovery from unexpected events.

We can observe that artificial intelligence brings many advantages in this case, but like any system, it also has its limitations. Currently, the main limitation of artificial intelligence is seen in its underdeveloped state, which does not provide complete autonomy to the project leader to focus on other tasks. The project leader still needs to use his knowledge to make decisions based on the results. The goal is to develop a system that does not require extensive human supervision, learns from the processes it has participated in, transfers experience to future projects, predicts risks, and makes independent decisions that the project team can trust. However, this also means that such a fully autonomous system would control the entire project, including stakeholder relationships and team management. Here, the future drawback of such a system becomes evident, as it would not possess the ability to empathize and connect with people, apply common sense reasoning when data doesn't make sense, employ communication tactics with clients, or act as a mentor to the project team as a human would. Additionally, such a system could also pose risks in terms of making incorrect decisions due to incorrect parameters input during programming, thereby compromising ethical, security, or data privacy standards. Questions arise regarding the legal responsibility for decisions made by the system and the potential for erroneous conclusions based on historical data. Based on the above, we conclude that artificial intelligence in project management is currently at a stage that allows

team members and project managers to save time, gain a good overview of project progress to identify potential deviations, and greatly facilitate communication and sending notifications through platforms among all stakeholders. Although these characteristics make it a valuable ally in the process, artificial intelligence still has limitations that prevent it from fully replacing human involvement and decision-making [55].

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